

Humidification

Final Lab Report

Unit Operations Lab 1

September 15, 2025

Group 4

Alexis Guillen

Kyle Hankosky

Khoa Nguyen

Rishabh Patel

1. Executive Summary

Heat and mass transfer coefficients for air molecules traveling through a wetted wall column were calculated using temperature data points from thermocouples, which were measured by using six different thermocouples placed at various locations throughout the setup. The experiment was conducted by alternating the air flow rates between 1, 4, and 6 SCFM, while maintaining a steady water flow rate of 0.163 GPM. The steam amount was slightly adjusted using a knob until the temperature readings stabilized for each trial. It was found that the temperature of the exiting air is at its highest and the humidity at its lowest when the air flow rate is 6 SCFM and the knob is fully open. Overall, this experiment was a success since the data obtained aligns with our expectations, making the calculations for heat and mass transfer coefficients straightforward.

2. Introduction

For this experiment, our main objective was to find the heat and mass transfer coefficients of air traveling through a wetted wall column. This is a very important experiment for us to understand, as concepts such as heat and mass transport are at the core of every chemical engineering process in industry. Within this experiment, a variety of apparatuses was used; a centrifugal pump was mainly used to transport water throughout the setup, a facility air compressor was used to transport air, a heat exchanger was used to provide heat to the setup, a regulator was used to set the air pressure, both the air and water rotameters were used to control the flows of their respective component, different thermocouples were used to measure the temperature of the air at various locations of the setup, an anemometer was used to measure the temperature and humidity of the exiting air, and valves were also used to allow both air and water to flow. To prepare for the experiment, we first turned on the air compressor and recorded the exiting air's temperature and humidity, the water reservoir was then filled with DI water and the water rotameter was set to a constant flow of 50% of 0.326 GPM throughout the experiment. The water pump and facility air compressor were then turned on for the experiment to start. The experiment was divided into 3 parts: room temperature, where there was no steam flow in the heat exchanger, moderate heat, where the steam knob was turned about halfway, and high heat, where the steam knob was fully turned. Within each part, we were also using 3 different flow rates of air: 1, 4, and 6 GPM. In total, we have 9 different trials to measure the temperature and humidity of air. In industry, humidification of air is used in textiles to prevent breakage and static buildup, woodworking shops use humidification systems to prevent cracking and expansion, and the printing industry also utilizes humidification systems to prevent paper curling. [1]

3. Theory

The thermodynamics concepts revolve around humidification describe heat and mass transfer coefficients by relating flux to driving force. For instance, $q = h \Delta T$ is used for heat transfer and $N_A = k_c \Delta C_A$ for mass transfer. The coefficients are also obtained from the relationship between dimensionless numbers such as Reynolds, Schmidt, and Sherwood numbers. In a wetted-wall column, mass transfer is governed by the differential balance $G'_s dY' = k_y (Y'_i - Y') P dz$, which can then be integrated to give us the number of transfer units (NTU) and the height of transfer unit (HTU). Similarly, heat transfer is governed by the equation $G'_s C'_s dY' = h (T_i - T) P dz$. It's also useful to note that heat and mass transfer can be independent of each other, and the Lewis relationship [2] gives us a useful connection between heat and mass transfer. From the measured humidity, we can convert to a mass ratio

using the equation $Y' = \frac{y}{1-y} \left(\frac{MW_{H_2O}}{MW_{Air}} \right)$, with y being the measured humidity. We can then

use Y' to calculate the number of transfer units $NTU = \ln \left(\frac{Y'_i - Y'_1}{Y'_i - Y'_2} \right)$. The mass transfer

coefficient k_y can then be calculated using the formula $k_y = \frac{NTU * Z * P}{G_s}$. Similarly, the heat

transfer coefficient h can also be calculated using the formula $h = \ln \left(\frac{T_i - T_1}{T_i - T_2} \right) \left(\frac{Z * P}{C_s - G_s} \right)$.

Using these equations, we can satisfy the experiment's main objective.

4. Results & Data analysis

The collected experimental data were reduced using the previous equations to calculate heat and mass transfer coefficients along with three dimensionless quantities that help us quantify the operating conditions within the column. When the mass transfer coefficients were plotted against the Reynold numbers (Figure 2), the only correlation observed was that higher temperatures caused larger differences between the mass transfer coefficients with no replicative trend. The behavior measured between the turbulence of the flow within the column and the boundary layer between the air and water did not show the expected outcome, which was that higher flow leads to a thinner interface between the two fluids. When plotted against the Reynolds number, the heat transfer coefficients (Figure 3) did show a trend in the data, which is that increased flow less efficient heat transfer between the two fluids due to the decreased contact time.

To compare the correlation given by the Sherwood equation (Figure 4), the data were first normalized and plotted against the Reynolds number on a base-ten log plot. A theoretical data set with points at regular intervals was then created to prove this correlation, showing that the data

collected vs the generated data set had a perfect correlation. This correlation shows that the humidity measurements taken and the interface temperatures measured are accurate for the process at hand.

The number of transfer units (NTU) was then evaluated with the air inlet temperature and flow rate as variables (Figure 5). For trials that had air flow rate as 4 and 6 SCFM, the data showed a positive correlation between inlet air temperature and NTUs. The outlier from this is the trials that used an air flow rate of 1 SCFM, which showed a negative correlation between the data. This result is indicative of better water vapor diffusivity at higher flows and temperatures, with the one SCFM trials likely reaching equilibrium lower in the column, reducing the gradient present compared to the higher flow rate trials.

Overall, the measured coefficients showed a range of trends, some consistent and some inconsistent with their theoretical correlations, although the inconsistencies in the trend do reflect experimental error in the operation of the experiment.

5. Discussion

Using relevant formulas described within the lab manual and Excel, we were able to obtain numerical values for the mass and heat transfer coefficients of all 9 trials. The results are organized in a table and are shown below. For clarification, trials 1-3 used no steam, trials 4-6 used a moderate amount of steam, and trials 7-9 used the maximum amount of steam.

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7	Trial 8	Trial 9
Mass Transfer Coefficient, k_y (kg/m ² /s)	0.002794	0.012689	0.02586	0.005351	0.009209	0.011753	0.003132	0.010799	0.010859
Heat Transfer Coefficient, h (J/m ² /s/K)	1.328366	5.55785	8.117479	5.391671	17.62025	23.9878	7.158103	22.5842	20.30887

Figure 1: Numerical Values for Mass and Heat Transfer Coefficients of All Trials

From this table, it is observed that trial 3 had the largest mass transfer coefficient value, with a value of 0.02586 kg/m²/s, while trial 6 had the largest heat transfer coefficient value out of all nine trials, with a value of 23.9878 J/m²/s/K. Trials 3, 6, and 9 used the largest amount air, so they would naturally have the highest mass transfer coefficients. However, the mass transfer coefficient for trial 3 is the largest when compared to trials 6 and 9, and this inconsistency could come from the sensitive knob of the air rotameter that is manually turned throughout the lab for controlling the air flow rate. In addition, when looking at the formula for mass transfer coefficient described in the theory section, we can see that it involves the mass ratio at the interaction interface between air and water in the column, which is rather unpredictable since the

water flows in a turbulent manner instead of laminar. On the other hand, trial 6 had the largest value for the heat transfer coefficient, despite using only a moderate amount of steam, while trial 9 used the largest amount of steam. Although bizarre, we must also compare the mass transfer coefficient for trials 6 and 9. From Figure 1, we can see that the mass transfer coefficient for trial 6 is greater than that of trial 9, with the trials having a mass transfer coefficient of 0.011753 kg/m²/s and 0.010859 kg/m²/s, respectively. So even though trial 6 used less amount of steam than trial 9, trial 6 carries more molecules of air through the water in the wetted column, allowing more heat to be transferred, which leads to a greater magnitude of the heat transfer coefficient.

6. References

<https://fogco.com/humidification-systems-benefit-various-industries/> [1]

<https://www.thermopedia.com/content/923/> [2]

Final Report Template Fall 2009, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.

CHE4162 Syllabus Spring 2010, Department of Chemical Engineering, Louisiana State University, Baton Rouge, LA.

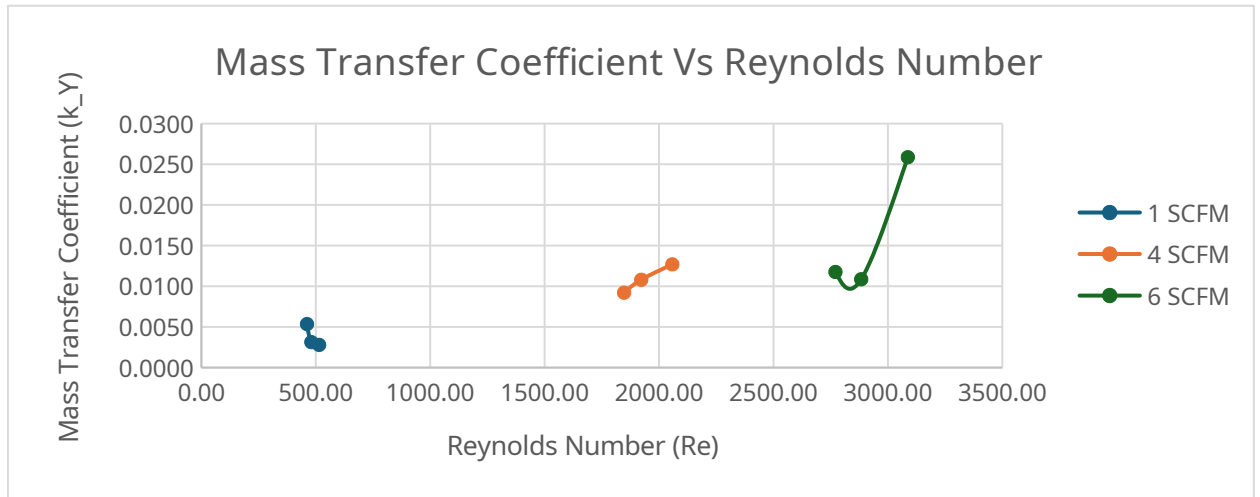
7. Appendix I: Data Tabulations/Graphs

Figure 22: Relationship Between Mass Transfer Coefficient and Reynolds Number with Three Air Flowrates

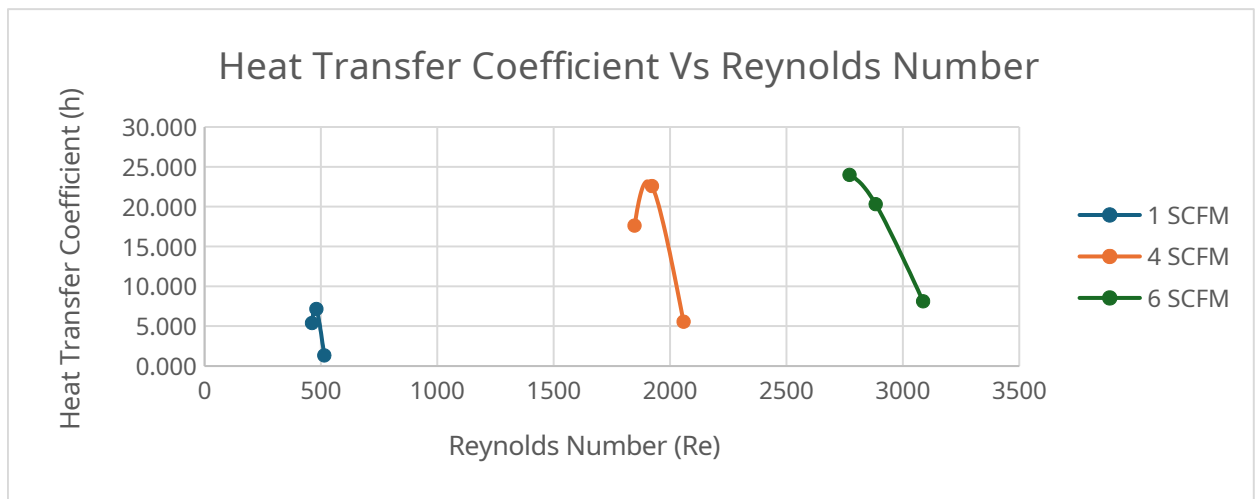


Figure 33: Relationship Between Heat Transfer Coefficients and Reynold Number with Three Air Flowrates

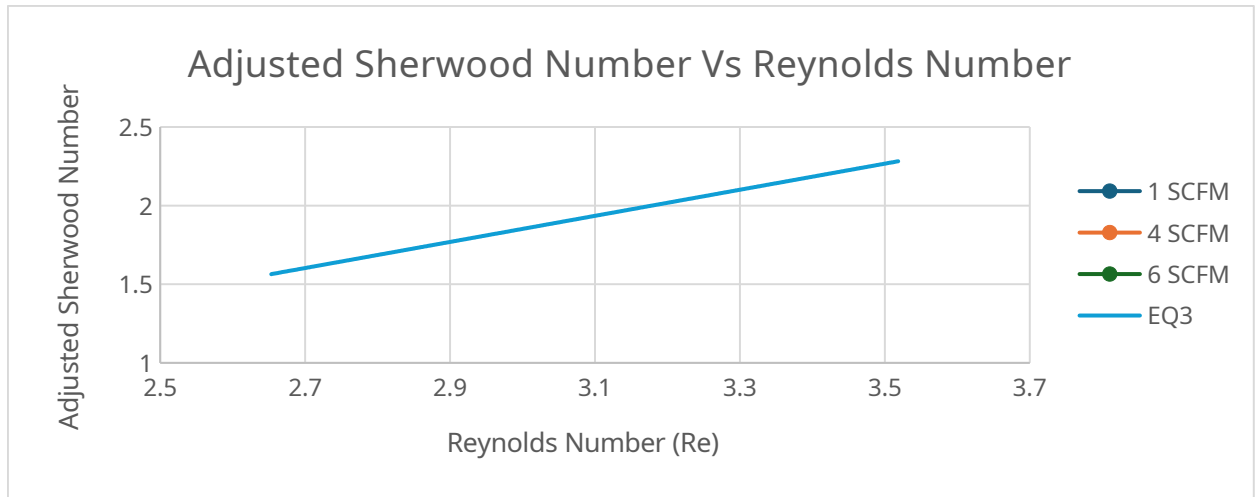


Figure 44: Relationship Between Sherwood Number and Reynold Number with Three Air Flowrates

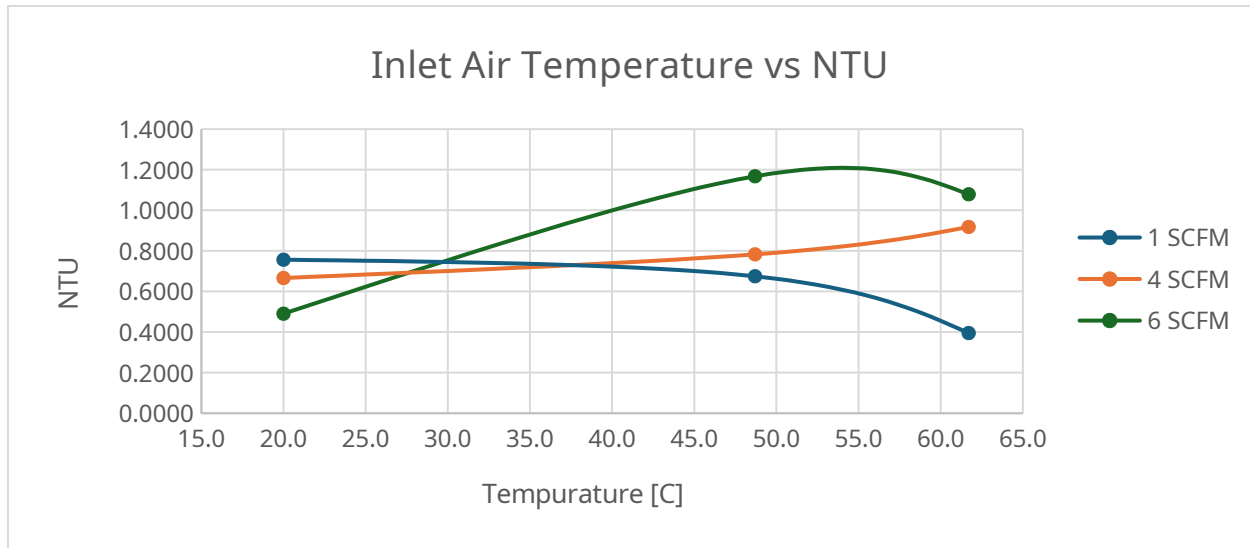


Figure 5:5 Relationship Between Air Inlet Temperatures and NTUs with Three Air Flowrates

NTU	
YP1	0.00095
YP2	0.01114
yi	0.03403
Ypi	0.02191
NTU	0.6661

Figure 6: Calculation of The Number of Theoretical Transfer Units using Mass Fractions

mass transfer	
circumference	0.24 m
Col Height	1.12 m
Yi	0.02191
Y1	0.00095
Y2	0.01114
Gs	0.00227
Ky	0.01269

Figure 8: Calculation of The Mass Transfer Coefficient Along With Humid Air

Heat transfer	
Psat	3.44851
Humidity Ratio WB	0.02191
Adjusted Humidity Ratio	0.02324
Ti	26.4333 C
T1	20 C
T2	23.1 C
Humid Head	1000.04
Mass flow	0.00227
perimeter	0.24 m
col height	1.12 m
Heat transfer	5.55785

Reynolds, Schmidt & Sherwood

Figure 7: Calculation of The Heat Transfer Coefficient Along with Humid air

rho	1.18
superficial V	0.42006
Viscosity	1.8E-05
Reynolds	2057.97
diffusion coefficient	2.5E-05
Schmidt	0.62077

Figure 9: Calculations of Reynolds, Schmidt, and Sherwood Number from Density, Cross Sectional Area, Superficial Velocity, and Diffusion Coefficient

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7	Trial 8	Trial 9
Air flowrate (SCFM)	1	4	6	1	4	6	1	4	6
Water flow rate (gpm)	0.163	0.163	0.163	0.163	0.163	0.163	0.163	0.163	0.163
Incoming air temp (C)	20	20	20	59	62	64	45	51	50
Initial exiting air temp (F)	70.8	70.8	70.8	72.1	72.1	72.1	72.1	72.1	72.1
Initial exiting air humidity (%)	12.6	12.6	12.6	12	12	12	12	12	12
Wet Bulb T1 (C)	22.8	23.1	22.9	31.8	33.6	34.9	28.8	30.8	32.7
Wet Bulb T2 (C)	23.8	24	23.9	38.7	42.4	44.9	33.3	38.4	38.9
Thermocouple 3 (C)	26.6	27.1	26.8	29.7	29.8	30.1	29.5	29.6	28.8
Thermocouple 4 (C)	25.9	26.3	26.1	29.1	29.2	29.4	29.1	29.1	28.3
Thermocouple 5 (C)	25.5	25.9	25.5	29.7	29.7	29.7	29.1	29.2	27.9
Thermocouple 6 (C)	25.5	25.3	24.8	30.2	30.1	29.9	29.5	29.5	28.7
Final exiting air temp (F)	74.4	72	74.8	96.6	103	108.6	88.3	95.3	100.7
Final exiting air humidity (%)	67.8	63.1	51.3	47	36.3	30	56.2	42.7	33.3
Room temp (F)	74.1	74.1	74.1	74.1	74.1	74.1	74.1	74.1	74.1
Room humidity (%)	59.2	59.2	59.2	59.2	59.2	59.2	59.2	59.2	59.2

Figure 10: Raw Experimental Data

8. Appendix II: Error Analysis

The largest source of quantitative error present in the data collected is in the temperature measurements required to calculate Y_1 , Y_2 and Y_i . It is likely that due to the temperature measurements being of the water at its interface with the air in the column, instead of the air at its interface with the water, caused inconsistencies to propagate. The NTU calculations are the most reflective of this error, due to the sensitivity of the calculation, slight variances in the temperatures within the column caused NTU values to widely differ from each other. This temperature uncertainty additionally propagated into the diffusivity calculation due to its reliance on the temperatures to calculate the mass transfer happening at the interface.

The largest sources of qualitative errors are as follows. First, uniform wetting of the column surface could not be guaranteed. And since the temperature measurements are in contact with this film, they likely disturbed the flow and led to inaccurate measurements. This would allow for the thickness of the film to differ probe to probe. Second, the assumption of a constant/steady-state interface temperature at the three points measured could also be a source of error. This is because the conditions within the column likely varied both radially and axially, causing a large variation in the calculated heat transfer coefficient. Lastly, no calibrations of measurement instruments were performed, and it was assumed to be accurate for humidity, temperature, and flow rate measurements. This may have caused an error due to the experiment being performed over two days, and with another team performing the experiment.

Finally, the error present in the correlations is more pronounced in higher inlet air temperatures, where we are much closer to the saturation point of the fluid. As a follow-up to the supposed calibration errors, small variations in the wet bulb measurements disproportionately affect the humidity ratio calculation, which could explain why the NTUs decreased with rising temperature in trials with the air flow rate of 1 SCFM.

9. Appendix III: Sample Calculations

- 1) The first thing that we need to calculate is the concentration of vapor in the gas in terms of mass ratio. It should be noted that in this set of calculations, we will be using data from trial 1 with the incoming air temperature of 20C and the air flowrate being 1 SCFM Now we must solve the following:

Y' = mass ratio at the transfer interface

Y_1 = mass ratio at the column inlet

Y_2 = mass ratio at the column outlet

The first step in this is the need to solve the mole fraction of water in the gas. But before we can do that, we need to find the saturation vapor pressure at the inlet and outlet of the column. We do this by using the following equation

$$P_{sat} = .6018 \left(\frac{17.27T_c}{T_c + 237.3} \right)$$

With T_c being the Temperature at the temperature gauge, which is 20 for Y_1 , the temperature at wet bulb 1 which was 22.8 Y_2 . Now when it comes to solving for Y' we use thermocouple 5 which was at a temperature of 25.5 Y' . We use thermocouple 5 due to it being closest to the incoming outlet and having the most interaction time with the air compared to the other thermocouples.

$$Y_1 P_{sat} = .6018 \left(\frac{17.27(20)}{20 + 237.3} \right) = 2.34 Pa$$

$$Y_2 P_{sat} = .6018 \left(\frac{17.27(22.8)}{22.8 + 237.3} \right) = 2.78 Pa$$

$$Y' P_{sat} = .6018 \left(\frac{17.27(25.5)}{25.5 + 237.3} \right) = 3.26 Pa$$

From here, we need to solve the actual vapor pressure. We do this by taking out saturated pressure and multiplying it by the humidity of each system.

$$Y_{1RH} = 2.34(0.066) = 0.154$$

The humidity we are using for the inlet is from letting the system run and giving us the baseline humidity.

$$Y_{2RH} = 2.78(0.678) = 1.88$$

The next step is solving the mole fraction for Y_1 and Y_2 . We find this by dividing by 1 standard atm in kPa

$$y_1 = \frac{0.154}{101.325} = 0.00152$$

$$y_2 = \frac{1.88}{101.325} = 0.0186$$

Now we can solve for the mass fraction of Y_1 and Y_2 using the following equation

$$Y' = \left(\frac{y_1}{1 - y_1} \right) \left(\frac{MW_{water}}{MW_{air}} \right)$$

Now plug in the values and solve for the mass fraction

$$Y_1 = \left(\frac{0.00152}{1 - 0.00152} \right) \left(\frac{18.02}{28.97} \right) = 0.000949$$

$$Y_2 = \left(\frac{0.0186}{1 - 0.0186} \right) \left(\frac{18.02}{28.97} \right) = 0.0118$$

Now we can properly solve for Y' . For this step since we don't have the humidity inside of the system we will need to solve it before we can continue. Since we already have saturated vapor pressure we can move onto finding the ratio at the wet bulb. Using the following equation.

$$62198 \left(\frac{P_{sat}}{101.325 - P_{sat}} \right)$$

Plugging in what we got for p_{sat} at Y_{psat}' we get a ratio of 0.0207. Using this value we can find the actual humidity ratio using

$$\frac{(2501 - 2.326 \cdot T) \cdot Ratio - 1.006 \cdot (T_1 \cdot T_2)}{(2501 + 1.86 \cdot T_1 - 4.186 \cdot T_2)}$$

Plugging in the values we have into the equation for the actual humidity ratio

$$\frac{(2501 - 2.326 \cdot 22.8) \cdot 0.0207 - 1.006 \cdot (20 \cdot 22.8)}{(2501 + 1.86 \cdot 20 - 4.186 \cdot 22.8)} = 0.0219$$

Now being able to solve for the mole fraction y_i at the column interface

$$y_i = \frac{Ratio}{0.62198 + Ratio}$$

$$y_i = \frac{0.0219}{0.62198 + 0.0219} = 0.0340$$

Now that we have the mole fraction, we can solve the mass ratio using the same equation as solving the previous two mass ratios.

$$Y' = \left(\frac{0.0340}{1 - 0.0340} \right) \left(\frac{18.02}{28.97} \right) = 0.0219$$

2) Number of Transfer Units

Now that we have all our mass ratios we can plug these into the following equation to get the number of transfer units.

$$Ntu = Ln \left(\frac{Y' - Y_1}{Y' - Y_2} \right)$$

$$Ntu = Ln \left(\frac{0.0219 - 0.000949}{0.0219 - 0.0118} \right) = 0.72695$$

3) Mass Transfer Coefficient

We have everything we need to solve for the mass transfer coefficient except for mass flow rate of dry gas which is solved for using the following:

$$Gs = 5.68 \cdot 10^{-4} \cdot Flow Rate$$

$$68 \cdot 10^{-4} \cdot 1 = 0.000568 \frac{Kg}{s}$$

Now we plug our knowns into this equation. The column height will be defined as $z=1.12m$ and circumference will be defined as $P = 0.24$

$$Ky = \frac{1}{Ntu \cdot \frac{z \cdot P}{Gs}}$$

$$\frac{1}{72695 \cdot \left(\frac{1.12 \cdot 0.24}{0.000568}\right)} = 0.002906 \frac{Kg}{m^2 s}$$

4) **Heat Transfer**

To solve for Heat transfer we use the following equation

$$h = Ntu \cdot \frac{Cs \cdot Gs}{z \cdot P}$$

Cs being the humid head which we would need to solve first before using the equation.

Cs = 100 + 1.86 · Actual Humidity ratio. Doing so we get the following Cs = 1000.04

$$\frac{Kj}{Kg \cdot K}$$

Plugging everything into the heat transfer equation we get the following

$$72695 \cdot \left(\frac{1000.04 \cdot 0.000568}{1.2 \cdot 0.24}\right) = 2481.128 \frac{J}{m^2 \cdot s \cdot K}$$

5) **Reynolds Number**

To solve for Reynolds number we need the following: Diameter of the column, density of vapor gas mix (ρ), superficial velocity (u), viscosity of air (μ). Solving for those variables, we can plug them into the equation

$$Re = \frac{\rho \cdot d \cdot u}{\mu}$$

$$\frac{18 \cdot 0.105 \cdot 0.0764}{0.0000184} = 514.49$$

6) **Schmidt's Number**

The following is the equation for Schmidt's Number, and we will only have to solve for the diffusion coefficient before we can solve for Schmidt's number.

$$Sc = \frac{\mu}{\rho \cdot D_{AB}}$$

Solving for the Diffusion Coefficient, we get $D_{AB} = 2.5 \cdot 10^{-5}$ Plugging in all the values, we get

$$\frac{0.0000184}{(1.84 \cdot 2.5 \cdot 10^{-5})} = 0.6223$$

7) **Sherwood Number**

Finally, we solve for the Sherwood equation.

$$Sh = 0.23Re^{0.83}Sc^{0.44}$$

$$0.23 \cdot 514.49^{0.83} \cdot 0.6223^{0.44} = 33.231$$

10. Appendix IV: Individual Team Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	All Team members
Pre-Lab Editing	1	All Team members
Experimental Objective (Prelab)	1	Rishabh Patel
Apparatus (prelab)	1	Khoa Nguyen
Materials & Supplies (prelab)	1	Khoa Nguyen
Procedure (prelab)	1	Alexis Guillen
Project Safety Evaluation (prelab)	1	Kyle Hankosky
Final Lab Editing	1	All Team members
Executive Summary (final lab)	2	Khoa Nguyen & Rishabh Patel
Introduction (Final lab)	2	Khoa Nguyen & Rishabh Patel
Theory (Final Lab)	3	Alexis Guillen & Khoa Nguyen
Results (final lab)	2	Kyle Hankosky & Alexis Guillen
Discussion (final lab)	2	Khoa Nguyen
References (final and/or prelab)	1	Rishabh Patel
Data Tabulation/Graphs (final)	12	Kyle Hankosky & Alexis Guillen
Error Analysis (final lab)	1	Kyle Hankosky
Sample Calculations (final Lab)	5	Kyle Hankosky & Alexis Guillen
Power Point Presentation	5	All Team members

Total	48 Hours	